



PRIORITY OCCUPATIONS AND PROMISING CREDENTIALS IN CYBERSECURITY

Cybersecurity Pathways:
A Guide to IT Careers and Certifications

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This document outlines a structured educational pathway for students interested in Information Technology and Cybersecurity. Starting in high school, it introduces students to a wide range of IT careers and includes foundational coursework in hardware, software, networking, and programming. The program emphasizes the importance of stackable credentials, work-based learning, and industry certifications. It provides a clear route from secondary to postsecondary education, enabling students to pursue advanced cybersecurity credentials and degrees. The document also covers career opportunities, job growth, and wage potential in the field.

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PROGRAMS OF STUDY



Equipping students with essential IT and Cybersecurity skills, this program prepares future leaders with valuable certifications and a clear path from high school to advanced postsecondary education.

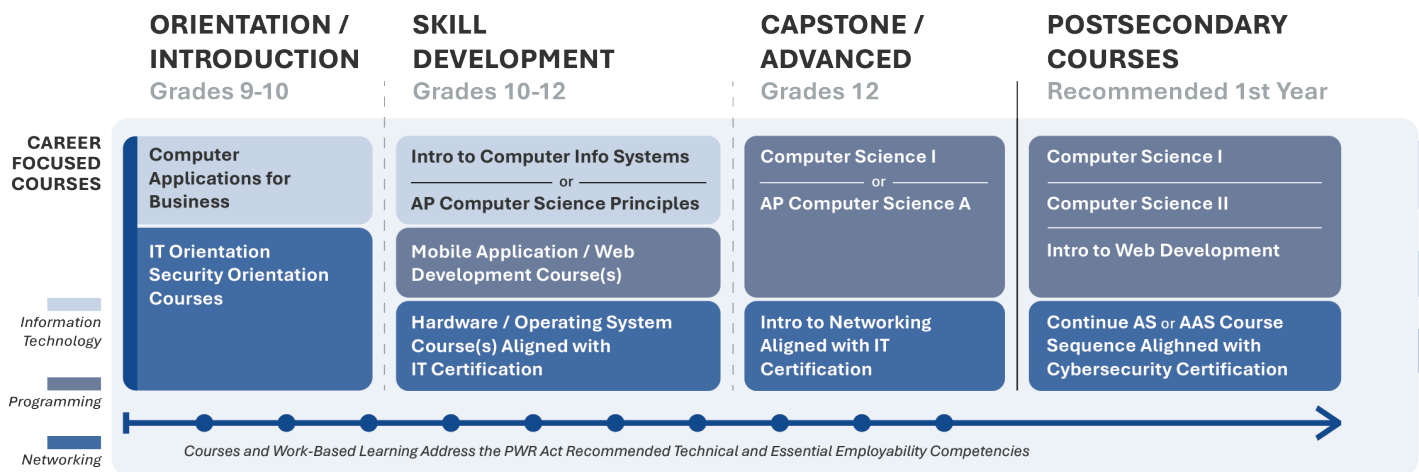
The Model Programs of Study for Cybersecurity is a subset of the Information Technology Model Programs of Study. It also begins in high school by introducing students to the broad range of careers in the Information Technology and Cybersecurity industry. Introductory and early college coursework combined with stackable credentials and work-based learning opportunities prepare students to demonstrate knowledge in fundamental IT areas like hardware, software, networking, and programming.

Students will have the opportunity to earn valuable industry certification through coursework at the secondary level. At the postsecondary level, students can pursue additional credentials in Cybersecurity while continuing to follow the Model Program of Study for Cybersecurity sequence through the bachelor's level.

OVERVIEW

HIGH SCHOOL CAREER-FOCUSED INSTRUCTIONAL SEQUENCE AND RELATED WORK-BASED LEARNING

Building the Future: Empowering Students with Industry Certifications in Cybersecurity and IT

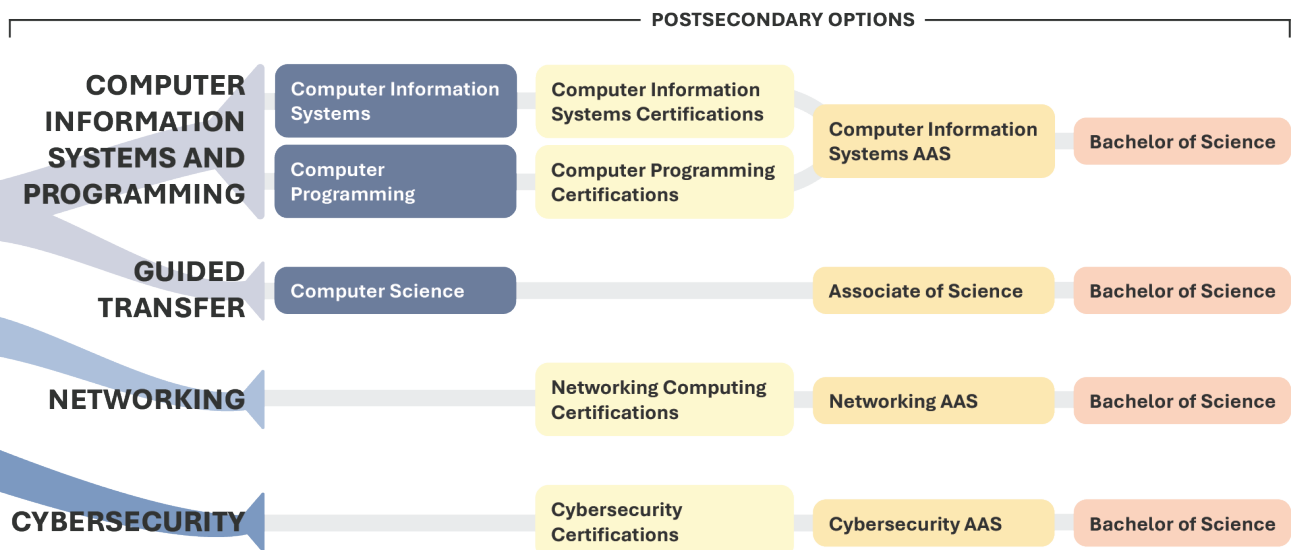


The high school career-focused instructional sequence in the Cybersecurity Model Programs of Study closely follows the sequence in the IT Model Programs of Study. The Orientation/Introduction courses include the IT orientation course but can also include a Security Orientation course. Students beginning this phase in their 9th grade year will maximize their opportunities for completing the secondary school sequence as well as earning industry certifications while in high school. During the Skill Development phase students should focus on building the basic computer skills including dual-credit and Illinois Articulation Initiative (IAI) opportunities in networking and computer programming courses.

Specific industry-oriented courses in computer hardware and operating systems courses further provide students with the opportunity to earn industry certification. During the Capstone/Advanced phase of the Cybersecurity Model Programs of Study students can begin to diverge into more cybersecurity-specific courses such as Introduction to Security or continue with IT networking courses. Students completing either the Introduction to Security or IT Networking courses would have the opportunity to earn an industry-recognized cybersecurity or networking certification based on which sequence they complete. Once students enter post-secondary education their educational tracks continue on a parallel pathway following closely the IT Model Programs of Study while adding cybersecurity courses throughout their studies.

MODEL PROGRAMS OF STUDY

The full Model Programs of Study for Cybersecurity can be used as a reference in local planning processes. The Guide then presents and describes in detail each component of the sequence, including descriptions of the underlying research, analysis, and Advisory Committee input leading to each component.



SELECTED OCCUPATIONS, WAGES, & JOB GROWTH

Program	Typical Job	Near or Above Living Wage Threshold for 1 Adult + 1 Child ¹	Median Hourly Wage ²	Growth in Illinois Annual Job Openings	Growth in Illinois: % Change Over 10 Years ²	Stackable?
1 Computer Information Sys. and Programming	Computer Systems Analysts	Y	\$41.67	2,230	9%	Typically Stacks to Bachelor's Program
	Computer User Support Specialists	Y	\$24.27	220	11%	
2 Computer Science	Computer and Information Systems Managers	Y	\$65.12	1,370	10%	
	Computer Hardware Engineers	Y	\$50.35	110	12%	
	Software Developers - Applications	Y	\$45.88	2,690	28%	
	Software Developers - Software	Y	\$51.63	1,030	13%	
	Computer and Information Research Scientists	Y	\$55.43	90	21%	
3 Networking	Computer Network Architects	Y	\$56.07	400	7%	
	Network and Computer Systems Administrators	Y	\$39.87	970	5%	
	Computer Network Support Specialists	Y	\$29.80	840	8%	
4 Cybersecurity	Information Security Analysts	Y	\$46.13	430	23%	
	Penetration & Vulnerability Tester	Y	\$41.12	N/A		
	Cybersecurity Manager	Y	\$53.00 (\$106,000/yr)	611	23%	
	Cybersecurity Architect	Y	\$57.50 (\$115,000/yr)	267	23%	
	Cybersecurity Specialist	Y	\$54.42 (\$108,852/yr)	393	23%	

1. <https://www.cyberseek.org/certifications.html>

2. <https://www.careeronestop.org/Toolkit/Careers/Occupations/occupation-profile.aspx?keyword=Penetration%20Testers&onetcode=15129904&location=Illinois>

STRATEGIC DUAL CREDIT COURSES: COMPETENCY STATEMENTS

COURSE DESCRIPTIONS

Each course description outlines the skills and knowledge students will gain, preparing them for IT careers and certifications.

ORIENTATION TO IT PROFESSIONS

This course enables students to analyze the field of information technology. The class will include a survey of the IT professions, employment skills, definitions, associations, current issues, salaries, and self-assessment survey of skills and competencies. This course will meet three times during the semester.

SECURITY AWARENESS

This course is intended to provide a basic survey of the importance of IT security awareness and data confidentiality. This security awareness-training course walks users through every aspect of information security in a very broad, easy to understand way and explains to them the value of securing data, both for themselves and the organization. The class will introduce legislation, local, state and federal privacy policies and liability of individuals and institutions related to data confidentiality and integrity. The course will introduce risk management, security policies, and common threats and countermeasures. The course will also present best practices in access control and password policies.

IT ESSENTIALS – A+

This course offers a hands-on approach to microcomputer maintenance. This course introduces a history of personal computer evolution. It examines the bus architecture, central processing unit parameters and replacement considerations, memory, video systems, storage devices and input/output devices. The course introduces the most popular and recent technologies. This course is designed to prepare the successful student for the CompTIA A+ hardware certification.

MANAGING IT – A+

This course offers a hands-on approach to managing information technology in a variety of environments. Students will be provided the skills needed to perform tasks such as installation, building, repairing, configuration, troubleshooting, optimizing, diagnosing and preventive maintenance in the context of the field service or enterprise environment and interacting with customers remotely. The course will also introduce critical responsibilities commonly required in managing IT within an organization including project management, help desk operations and information security. This course is aligned to the CompTIA A+ certification.

NETWORK ESSENTIALS

This course provides an introduction to the operation of computer networks and networking devices. The course also provides an examination of the history and evolution of data communications and computer networks. Hardware and software network configurations, operations and requirements will be discussed. Topics include network media and topologies, the OSI model, protocols, standards, technologies, network implementation, and network support. The course content aligns to some of the CompTIA Network+ objectives.

NETWORK SERVICES

This course is a continuation of Network Essentials with an emphasis on network management, analysis, and addressing. The course will introduce essential network services such as ftp, http, dnsc, dns, messaging, authentication and wins. The course provides hands-on exercises in which each student will be required to configure network services. Emphasis will be on configuration, analysis, proactive maintenance, and troubleshooting. This course is designed to supplement the A+, Network+ and Server+ certifications.

IT SECURITY ESSENTIALS – SECURITY+

This course offers a comprehensive introduction to network security, guiding students through the essential principles and practices in the field. Students will learn about computer network vulnerabilities and threats, as well as how to identify and mitigate them. The curriculum covers critical topics such as network security planning, the latest security technologies, and the structure and organization of security systems within an enterprise. Additionally, the course addresses the legal and ethical issues that professionals may encounter in network security. Designed to align with the CompTIA Security+ certification, this course equips students with the knowledge and skills necessary for a career in cybersecurity.

NICE FRAMEWORK: CYBERSECURITY COMPETENCY AREAS

Competency Area Name	Competency Area Description
Access Controls	This Competency Area describes a learner's capabilities to define, manage, and monitor the roles and secure access privileges of who is authorized to access protected data and resources and understand the impact of different types of access controls.
Artificial Intelligence (AI) Security	This Competency Area describes a learner's capabilities to secure Artificial Intelligence (AI) against cyberattacks, to ensure it is adequately contained where it is used, and to mitigate the threat AI presents where it or its users have malicious intent.
Asset Management	This Competency Area describes a learner's capabilities to conduct and maintain an accurate inventory of all digital assets, to include identifying, developing, operating, maintaining, upgrading, and disposing of assets.
Cloud Security	This Competency Area describes a learner's capabilities to protect cloud data, applications, and infrastructure from internal and external threats.
Communications Security	This Competency Area describes a learner's capabilities to secure the transmissions, broadcasting, switching, control, and operation of communications and related network infrastructures.
Cryptography	This Competency Area describes a learner's capabilities to transform data using cryptographic processes to ensure it can only be read by the person who is authorized to access it.
Cyber Resiliency	This Competency Area describes a learner's capability related to architecting, designing, developing, implementing, and maintaining the trustworthiness of systems that use or are enabled by cyber resources in order to anticipate, withstand, recover from, and adapt to adverse conditions, stresses, attacks, or compromises that use or are enabled by cyber resources.
DevSecOps	This Competency Area describes a learner's capabilities to integrate security as a shared responsibility throughout the development, security, and operations (DevSecOps) life cycle of technologies.
Operating Systems (OS) Security	This Competency Area describes a learner's capabilities to install, administer, troubleshoot, backup, and conduct recovery of Operating Systems (OS), including in simulated environments.
Operational Technology (OT) Security	This Competency Area describes a learner's capabilities to improve and maintain the security of Operational Technology (OT) systems while addressing their unique performance, reliability, and safety requirements.
Supply Chain Security	This Competency Area describes a learner's capabilities to analyze and control digital and physical risks presented by technology products or services purchased from parties outside your organization.

June 13, 2024 | <https://niccs.cisa.gov/workforce-development/nice-framework/competency-area>

NIST/NICE FRAMEWORK COMPETENCIES FOR CYBERSECURITY WORK

Competency Area Name	Competency Type	Competency Area Description
Asset and Inventory Management	Organizational	This Competency describes a learner's capabilities related to the process of identifying, developing, operating, maintaining, upgrading, and disposing of assets.
Business Acumen	Organizational	This Competency describes a learner's understanding of key business issues and the strategic management of human, financial, and information resources in order to achieve desired organizational outcomes. Includes taking into account internal and external influences (e.g., business, economic, political, social trends) in decision making.
Business Continuity	Organizational	This Competency describes a learner's capabilities related to business continuity planning and continuity of operations to help ensure an organization can prevent and overcome serious incidents or disasters and quickly resume normal operations within a reasonable timeframe.
Collection Operations	Technical	This Competency describes a learner's capabilities related to executing the collection of cybersecurity information to develop intelligence using appropriate strategies and within the priorities established through the collection management process. This Competency is specific to federal work related to Title 10 and 50 of the U.S. Code.
Communication	Professional	This Competency describes a learner's capabilities related to the process of clearly and effectively expressing information or ideas to individuals or groups in a variety of ways (verbal, nonverbal, written, and visual). Includes understanding when and how to adapt messages for different audiences as well as listening to others' instructions, ideas and intentions, attending nonverbal cues, and responding appropriately.
Computer Languages	Technical	This Competency describes a learner's capabilities related to computer languages and their applications in order to enable a system to perform specific functions.
Conflict Management	Professional	This Competency describes a learner's capabilities related to managing and resolving conflicts, grievances, confrontations, or disagreements in a constructive manner to minimize negative personal impact and collaborating with others to encourage cooperation and teaming.
Contracting and Procurement	Organizational	This Competency describes a learner's capabilities related to procuring, negotiating, administering, and managing various types of contracts, including application of contracting or procurement techniques and requirements according to applicable laws and policies.

Competency Area Name	Competency Type	Competency Area Description
Critical Thinking	Professional	This Competency describes a learner's capabilities related to analysis, reasoning, problem solving, planning, and organization to make inferences or draw conclusions that enable sound decision-making in their work.
Data Analysis	Technical	This Competency describes a learner's capabilities related to systematically applying statistical and logical techniques to describe, illustrate, condense, summarize, and evaluate data. The synthesis and analysis of various types of data to reach a decision, make a recommendation, or to compile reports, briefings, executive summaries, and other correspondence to support organizational work, goals, and plans.
Data Management	Organizational	This Competency describes a learner's capabilities related to the development and execution of data management plans, programs, practices, processes, architectures, and tools across all stages of the data lifecycle. Includes processes around the creation, storage, archiving, discovery, access, disposal, enhancement, and reuse of data and information assets.
Data Privacy	Organizational	This Competency describes a learner's capabilities related to the proper handling of data to protect individuals' privacy, including addressing whether and how data is shared and with whom; how data is collected and stored; and legal and regulatory compliance.
Data Security	Technical	This Competency describes a learner's capabilities related to the methods and procedures that protect data and information systems by ensuring their confidentiality, integrity, and availability.
Database Administration	Technical	This Competency describes a learner's capabilities related to principles, methods, and tools for managing and maintaining database systems used to store and organize data to maximize the value, quality, and usability of data resources. May include the development and enforcement of database standards, guidelines, and operational policies and procedures.
Digital Forensics	Technical	This Competency describes a learner's capabilities related to the application of tools and techniques used in data recovery and preservation of electronic evidence. Includes the collection, processing, preservation, analysis, and presentation of computer-related evidence in support of network vulnerability mitigation and criminal, fraud, counterintelligence, or law enforcement investigations.
Education and Training Delivery	Organizational	This Competency describes a Learner's capabilities related to helping others acquire knowledge or develop skills. Includes training of personnel within pertinent subject domain, including cybersecurity awareness for users of computers and other electronic telecommunication devices.

Competency Area Name	Competency Type	Competency Area Description
Education and Training Curriculum Development	Organizational	This Competency describes a Learner's capabilities related to developing, planning, coordination, delivery, and evaluation of training courses, methods, and techniques.
Encryption	Technical	This Competency describes a learner's capabilities related to the cryptographic process of transforming data to ensure that it can only be read by the person who is authorized to access it.
Enterprise Architecture	Technical	This Competency describes a learner's capabilities related to the principles, concepts, and methods of enterprise architecture to align technology strategy, plans, and systems with the mission, goals, structure, and processes of the organization.
Identity Management	Technical	This Competency describes a learner's capabilities related to the tools, policies, and systems that define, manage, and monitor the roles and secure access privileges of users, ensuring authorized people and things have the appropriate access to the correct protected resources at the correct time.
Incident Management	Technical	This Competency describes a learner's capabilities related to the tactics, technologies, principles, and processes to analyze, prioritize, and handle cybersecurity incidents.
Information Systems and Network Security	Technical	This Competency describes a learner's capabilities related to the methods, tools, and procedures—including development of information security plans—to detect, respond, and protect information, information systems, and networks from risks and to provide or restore security of information systems and network services.
Information Technology Assessment	Technical	This Competency describes a learner's capabilities related to the principles, methods, and tools (for example, surveys, system performance measures) to assess the effectiveness and practicality of technology systems.
Infrastructure Design	Technical	This Competency describes a learner's capabilities related to the architecture and topology of software, hardware, and networks, including LANS, WANS, and telecommunications systems, their components and associated protocols and standards, and how they operate and integrate with one another and with associated controlling software.
Intelligence Analysis	Technical	This Competency describes a learner's capabilities related to the application of individual and collective cognitive methods to collect, process, and interpret data about an enemy to answer tactical questions about current operations or to predict future behavior.

Competency Area Name	Competency Type	Competency Area Description
Interpersonal Skills	Professional	This Competency describes a learner's capabilities related to developing and maintaining relationships with others in order to work effectively. Includes being sensitive to and inclusive of cultural diversity, race, gender, disabilities, and other individual differences as well as considering and responding appropriately to the needs, feelings, and capabilities of customers and colleagues (subordinates, peers, and superiors).
Knowledge Management	Organizational	This Competency describes a learner's capabilities related to the management and administration of processes and tools that enable the organization to identify, document, and access intellectual capital and information content.
Law, Policy, and Ethics	Organizational	This Competency describes a learner's capabilities related to awareness of and compliance with laws, regulations, policies, and ethics that can impact organizational activities.
Mathematical Reasoning	Technical	This Competency describes a learner's capabilities related to solving practical problems and determining if an assertion is correct by choosing appropriately from a variety of mathematical and statistical techniques.
Modeling and Simulation	Technical	This Competency describes a learner's capabilities related to mathematical modeling and simulation tools and techniques to plan and conduct test and evaluation programs, characterize systems support decisions involving requirements, evaluate design alternatives, or support operational preparation.
Network Management	Technical	This Competency describes a learner's capabilities related to the operation, management, and maintenance of computer network and telecommunication systems and linked systems and peripherals.
Operating Systems	Technical	This Competency describes a learner's capabilities related to computer operating systems.
Operations Support	Technical	This Competency describes a learner's capabilities related to the policies and procedures used to ensure the production or delivery of products and services, including tools and mechanisms for distributing new or enhanced hardware and software.
Organizational Awareness	Organizational	This Competency describes a learner's capabilities related to understanding an organization's mission and functions; its social and political structure; and how programs, policies, procedures, rules, and regulations drive and impact the work and objectives of the organization.
Physical Device Security	Technical	This Competency describes a learner's capabilities related to building, maintaining, repairing, and securing electronic devices, associated peripherals, and accessories.

Competency Area Name	Competency Type	Competency Area Description
Policy Development	Leadership	This Competency describes a learner's capabilities related to the process of creating, communicating, and maintaining policies and procedures within an organization.
Problem Solving	Professional	This Competency describes a learner's capabilities related to determining the accuracy and relevance of information; using sound judgment to generate and evaluate alternatives; and making well-informed, objective recommendations and decisions that take into account facts, goals, constraints, and risks while perceiving the impact and implications of the decisions.
Process Control	Organizational	This Competency describes a learner's capabilities related to the active changing of the process based on the results of process monitoring .
Project Management	Organizational	This Competency describes a learner's capabilities related to the principles, methods, and tools for developing, scheduling, coordinating, and managing projects and resources, including monitoring and inspecting costs, work, and contractor performance.
Requirements Analysis	Technical	This Competency describes a learner's capabilities related to the principles and methods to identify, analyze, specify, design, and manage functional and infrastructure requirements—includes translating functional requirements into technical requirements used for logical design or presenting alternative technologies or approaches.
Risk Management	Organizational	This Competency describes a learner's capabilities related to the principles, methods, and tools used for risk assessment and mitigation, including assessment of failures and their consequences. Includes the oversight, evaluation, and support of the documentation, validation, assessment, and authorization processes necessary to assure that existing and new technology systems meet organizational cybersecurity and risk requirements.
Software Development	Technical	This Competency describes a learner's capabilities related to the principles, methods, and tools for designing, developing, modifying and testing software in a given environment, embodying all the stages throughout the software development lifecycle.
Software Testing and Evaluation	Technical	This Competency describes a learner's capabilities related to the principles, methods, and tools for developing, administering, and analyzing software testing to evaluate, verify, and validate technical, functional, and performance characteristics (including interoperability).

Competency Area Name	Competency Type	Competency Area Description
Strategic Relationship Management	Organizational	This Competency describes a learner's capabilities related to the concepts, practices, and techniques used to identify, engage, influence, and monitor relationships with individuals and groups connected to a work effort—including those actively involved, those who exert influence over the process and its results, and those who have a vested interest in the outcome (positive or negative)—to build strategic relationships and achieve common goals.
Strategic Planning	Leadership	This Competency describes a learner's capabilities related to formulating effective tactics and metrics associated with the vision, mission, goals, and objectives of the organization or business unit.
Supply Chain Management	Organizational	This Competency describes a learner's capabilities related to the process of analyzing and controlling risks presented to your company, data, operations and finances through the products or services purchased from parties other than your own company.
System Administration	Technical	This Competency describes a learner's capabilities related to the installation, configuration, troubleshooting, and maintenance of computer systems to ensure their confidentiality, integrity, and availability. Includes the management of accounts, firewalls, and patches as well as access control, passwords, and account creation and administration.
Systems Integration	Technical	This Competency describes a learner's capabilities related to the principles, methods, and procedures for installing, integrating, and optimizing information systems components.
Systems Testing and Evaluation	Technical	This Competency describes a learner's capabilities related to the principles, methods, and tools for developing, administering, and analyzing systems test procedures to evaluate, verify, and validate technical, functional, and performance characteristics (including interoperability) of systems or elements of systems incorporating technology. Includes the identification of critical operational issues.
Target Development	Technical	This Competency describes a learner's capabilities related to the systematic examination of potential target systems - and their components, individual targets, and even elements of targets - to determine the necessary type and duration of the action that must be exerted on each target to create an effect that is consistent with the commander's specific objectives.
Technology Fluency	Technical	This Competency describes a learner's capabilities related to keeping up to date on technological developments, including new applications of technology, emerging technologies, and the effective use of technology to achieve results.

Competency Area Name	Competency Type	Competency Area Description
Telecommunications	Technical	This Competency describes a learner's capabilities related to the transmissions, broadcasting, switching, control, and operation of telecommunications and related network infrastructure.
Threat Analysis	Technical	This Competency describes a learner's capabilities related to the identification and assessment of adversaries' or foreign intelligence entities' cybersecurity attack capabilities and activities. Includes awareness and evaluation of internal and external information vulnerabilities that may be at risk to cyber-attacks and the production of findings to help initialize or support law enforcement and counterintelligence investigations or activities.
Vulnerabilities Assessment	Technical	This Competency describes a learner's capabilities related to the principles, methods, and tools for assessing vulnerabilities and developing or recommending appropriate mitigation countermeasures.
Workforce Management	Leadership	This Competency describes a learner's capabilities related to building and managing a productive workforce based on organizational goals, budget considerations, and staffing needs. Ensures that employees are appropriately recruited, selected, appraised, and rewarded; takes action to address performance problems. Includes retention and career pathing strategies (including mentoring or coaching) for a variety of work situations.

June 2022 | <https://csrc.nist.gov/News/2023/nice-framework-competencies-nist-ir-8355>





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